Grade 6 Geography

Solved Exercises

Complete Book - All 6 Chapters

Balochistan Textbook Board (BTBB)

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Chapter 1: Structure of Earth and Types of Rocks

Exercise (pages 17-18): 1. Choose the correct answer

I. The type of rock form through cooling and solidification of magma or lava is

- Sedimentary rock
- Metamorphic rock
- ✓ Igneous rock
- none of the above

II. Sedimentary rocks are formed by:

- heat and pressure
- ✓ erosion and deposition
- volcanic activity
- none of the above

III. The idea of plate tectonic was first proposed by

- ✓ Alfred Wegener
- Isaac Newton
- Galileo Galilei
- Aristotle

IV. Rocks that have changed their properties due to pressure and temperature are called

- Igneous rocks
- Sedimentary rocks

✓ Metamorphic rocks

none of these

Exercise: 2. Fill in the blanks:

i. Igneous rocks are formed from the cooling of magma or ____.

lava

ii. Sedimentary rocks are made up of ____ that accumulate over time.

sediments

iii. ____ rocks undergo changes due to heat, pressure, or chemical reactions.

Metamorphic

iv. Geography is the combination of two ____ words.

Greek

v. The outermost layer of the earth is called ____.

crust

Note: the printed exercise misnumbers the last two fill items (the fourth is printed as vi. and the fifth as V.); they are given here in corrected sequential order.

Exercise: 3. Write short answers to the following questions:

I. Define rock.

A rock is any naturally occurring, solid, non-living material made up of minerals or mineral-like substances. The outer layer of the earth

is made up of different rocks, which are classified by how they were formed.

II. What is the difference between intrusive and extrusive igneous rocks?

Intrusive igneous rocks form when magma cools slowly inside the earth, so they have large crystals; examples are granite and diorite. Extrusive igneous rocks form when lava cools quickly on the surface of the earth, so their crystals are small and few; examples are andesite and basalt.

III. How are sedimentary rocks formed? Give an example.

Wind, water, heat and ice break igneous and metamorphic rocks into small particles. These particles are carried away and deposited layer upon layer in low-lying water bodies, where pressure from above and cementing material turn them into solid rock. Examples are limestone, sandstone and shale.

IV. Write names of major tectonic plates in the world.

The seven major tectonic plates named in the chapter are the Pacific Plate, the African Plate, the North American Plate, the Eurasian Plate, the South American Plate, the Antarctic Plate and the Indo-Australian Plate.

Exercise: 4. Answer the following questions in detail:

I. Write a note on internal structure of earth.

The earth has three layers. The crust is the outermost layer on which we live; it is made of solid rocks, with the oceanic crust 5 to 7

kilometres thick and the continental crust 25 to 70 kilometres thick. The mantle lies below the crust and is about 2900 kilometres deep; its upper molten layer, the asthenosphere, helps the upper layer to move. The core is the innermost layer: the outer core, from 2900 to 5150 kilometres, is molten, while the inner core, from 5150 to 6370 kilometres, is solid because of the immense pressure and consists mostly of iron and nickel.

II. Explain igneous rocks in detail.

Igneous rocks are formed from molten material. When magma inside the earth, or lava flowing on the surface, cools and solidifies, it becomes igneous rock. There are two main kinds: intrusive igneous rocks, which cool slowly inside the earth, and extrusive igneous rocks, which cool quickly on the surface. Igneous rocks have no layers, they are very hard and dense, and no fossils are found in them.

III. Explain the rock cycle, including the processes involved and how different rock types are transformed into one another.

The rock cycle is a continuous process. First, molten magma cools and forms igneous rock. Then weathering breaks this rock into small particles, which are transported and deposited layer upon layer and, under pressure, become sedimentary rock. When sedimentary rock is buried deep in the earth, heat and pressure change its colour and chemical composition and it becomes metamorphic rock. With still more heat it melts again into magma, and when the magma cools, igneous rock forms again; so the cycle keeps repeating.

IV. Discuss modern techniques in the field of geography.

Four modern techniques are described in the chapter. GIS, the Geographic Information System, is a computer system that stores information about places in layers, analyses it and presents it as maps. GPS, the Global Positioning System, is a satellite system that lets a receiver find its position anywhere on earth where there is an unobstructed line of sight to four or more satellites. Google Maps uses such data for navigation and finding places. Remote sensing collects information about distant or inaccessible places with special cameras carried by aircraft and satellites, and GPS guidance also lets modern unmanned aircraft (drones) reach their targets.

Activity (page 18) - Guidance

Guidance: collect various rock samples (igneous, sedimentary and metamorphic) or use high-quality images, label each sample with a number or letter, and create a worksheet or quiz in which students match the rock type to its sample and discuss the characteristics of each type. Also take a short walk around the school or nearby area, collect small rocks and discuss how they could be used. This is a practical activity; the book fixes no single written answer for it.

Chapter 2: Mountains, Plateaus and Valleys

Exercise (page 34): 1. Select the correct answers:

I. Rockies are located in

- Asia
- Europe
- South America
- ✓ North America

II. The highest mountain range in the world is

- Alps
- ✓ Himalayas
- Rockies
- Andes

III. The highest mountain peak in the world is

- K2
- ✓ Mount Everest
- Nanga Parbat
- Mount Kilimanjaro

IV. The highest mountain peak in Pakistan is

- Mount Everest
- Nanga Parbat
- ✓ K2
- Annapurna

V. The process primarily forms U-shaped valleys is

River erosion

✓ Glacial erosion

Wind erosion

Volcanic activity

Exercise: 2. Fill in the blanks:

I. Andes is a famous mountain range in ____.

South America

ii. V-shaped valleys are primarily formed by ____ erosion over long periods.

river

iii. ____ valleys are formed by the erosive action of glaciers.

U-shaped

iv. The Himalayas form the ____ boundary of the subcontinent.

northern

v. Ras Koh is situated in ____.

Balochistan

Note: the printed exercise misnumbers the fourth fill item as vi. (before item v.); the items are given here in corrected sequential order.

Exercise: 3. Answer the following short questions:

I. Define mountain.

A mountain is a part of the earth that is high and sloping compared with the surrounding surface. Mountains are outstanding landforms that shape the surface of the earth, and they are formed in different ways, including tectonic movements of the earth and volcanic activity.

II. Write the names of famous mountain ranges in Pakistan.

The famous mountain ranges of Pakistan are the Himalayas, the Karakoram and the Hindu Kush. Other ranges described in the chapter are the Koh-e-Sufaid, the Koh-e-Sulaiman and the Kirthar Range, and in Balochistan the Toba Kakari Range and the Ras Koh Range.

III. What is meant by V-shaped valley?

A V-shaped valley is a valley formed by river erosion whose shape is like the English letter V. The river cuts into the surface over which it flows and carries the broken material to low-lying areas, so the valley keeps getting deeper.

IV. What are different types of plateaus?

There are three types of plateaus: the intermountain plateau, which is surrounded on all sides by high mountain ranges; the piedmont plateau, which lies at the foot of high mountains; and the continental plateau, which has plains or sea around it and lies far from mountain ranges.

Exercise: 4. Answer the following questions in detail:

I. What are different types of plains? Elaborate.

Plains are classified by how they were formed into three kinds. A structural plain is formed when internal movements of the earth raise a wide area whose surface is flat; examples are the Great Plains of North America and the steppes of Asia and Eastern Europe. An erosional plain is formed when heat, ice, water and wind wear down high areas and carry the material elsewhere; examples are the plains of northern Canada. A depositional plain is formed when material carried by rivers, water, sea and glaciers is deposited layer upon layer.

II. Explain different types of plateaus.

There are three types of plateaus. An intermountain plateau is surrounded on all sides by high mountain ranges, with high mountains on all four sides; the best example is the Tibetan Plateau. A piedmont plateau lies at the foot of mountains, with a mountain on one side and a plain or sea on the other; the best example is the Patagonia Plateau in southern Argentina. A continental plateau is a plateau around which there are plains or sea and which lies far from mountain ranges; the best example is the Arabian Plateau.

III. How V-shaped and U-shaped valleys are formed?

A V-shaped valley is formed by river erosion: the river cuts the surface over which it flows, carries the broken material downstream and deepens its channel, so the valley takes the shape of the English letter V. A U-shaped valley is formed by a glacier: large masses of ice moving down a slope are called glaciers, and when a

glacier moves through a river valley that already exists, its enormous weight breaks up the ground ahead of it and carries it along; after the glacier has passed, the valley takes a U shape, which is wider than a V-shaped valley.

IV. Write a note on the importance of Himalayas in Pakistan.

The northern mountain ranges are important for Pakistan in several ways. They lie in the north and west of Pakistan and therefore act as natural northern and western boundaries. In the rainy season the moisture-laden monsoon winds coming from the Indian Ocean strike them, rise and cause rainfall. All the rivers of Pakistan come through these mountains into the plains, and the snow melting in them adds water to the rivers, on which the prosperity of Pakistan depends. The highest peaks of the world lie in these mountains, which climbers from foreign countries visit every year, and the beautiful lakes and valleys inside them are very attractive for tourists. They also stop the cold winds from Siberia from entering Pakistan in winter, and valuable stones, plants, animals and different minerals are found in them.

Activity (page 34) - Guidance

Guidance: use maps to identify different landforms and make a list of the landforms surrounding you. This is a practical activity; the book fixes no single written answer for it.

Chapter 3: Climatic Regions of the World

Exercise (pages 45-46): 1. Choose the correct answer

I. Which of the following best describes weather?

- long-term average conditions
- consistent patterns over 30 years
- ✓ short-term atmospheric state
- slow changes over large regions

II. An example of weather is

- monsoon climate
- Sahara Desert
- ✓ daily temperature fluctuations
- consistent rainfall patterns

III. The term refers to average weather patterns over a longer period is

- ✓ climate
- meteorology
- tornadoes
- thunderstorms

IV. The amount of solar energy received at a location determines

- altitude
- ✓ distance from the equator
- ocean currents

land shape (relief)

V. The zone experiences extremely cold temperature near the poles is

tropical

temperate



polar

arid

Exercise: 2. Mark the following statements either true or false:

I. Weather refers to long-term atmospheric conditions. False

II. Climate changes rapidly and varies over short distances. False

III. Climate describes average weather patterns over a longer period. True

IV. Weather is consistent over larger regions. False

Note: the first statement is false because long-term average conditions are climate, not weather. The second is false because rapid change over short distances is a property of weather. The fourth is false because weather is local and short-term, while consistency over large regions is a property of climate.

Exercise: 3. Answer the following short questions:

I. Define weather and climate.

Weather is the atmospheric condition of a small place for a short time. Its elements are temperature, rainfall, wind speed, clouds,

humidity and air pressure, and these elements keep changing all the time. Climate is the average condition of each weather element when it is fixed for a wide area and for many years; at least thirty years of weather information is needed to fix a climate, and climate does not change as quickly as weather.

II. How do ocean currents influence climate?

Water flowing in the seas and lakes throughout the year is called an ocean current, and cold and warm currents affect the climate around them because of their cold or warm character. For example, the warm Gulf Stream reaches western Europe and makes its climate warm, while the cold California current makes the climate of the west coast of North America cold.

III. Why are higher altitudes cooler?

Higher areas are colder because, as one goes upward from sea level, the air becomes less dense and its capacity to absorb the heat of the sun decreases, so the temperature falls. That is why the climate of areas high above sea level is less warm and cold.

IV. What are the two vital reasons for the importance of monsoons?

Two vital reasons for the importance of monsoons are: first, the summer monsoon rains supply water to the rivers, and second, these rains benefit agriculture because the crops get water. The chapter also notes that sometimes these rains can cause floods.

Exercise: 4. Answer the following questions in detail:

I. Describe the factors that affect the climate of any place.

Six factors affect the climate of a place. First, latitude: the rays of the sun fall vertically on the equator all year, so its climate is warm, and as one moves towards the poles the rays fall more slantingly, so the climate becomes less warm and colder. Second, altitude above sea level: with height the air becomes less dense and absorbs less heat, so high areas are colder. Third, ocean currents: warm currents warm the nearby climate and cold currents cool it. Fourth, topography: mountains block the pressure of the wind and make one side moist and the other side dry, creating dry regions. Fifth, the distribution of land and sea: land areas heat and cool quickly, while water areas heat and cool slowly. Sixth, prevailing winds: winds coming from cold areas are cold, from warm areas are warm, from the sea are moist, and from deserts are hot and dry.

II. Discuss tropical zone in detail.

The tropical zone lies on both sides of the equator, between the Tropic of Cancer (23.5 degrees north latitude) and the Tropic of Capricorn (23.5 degrees south latitude). It is very hot throughout the year, and its average annual temperature is more than 26 degrees Celsius (78.8 degrees Fahrenheit). Rainfall is heavy, with the highest annual rainfall exceeding 1500 millimetres (59 inches). Rainforests grow here, containing more than half of the plant and animal species of the world, and the Amazon rainforests are the largest tropical rainforests in the world. Even so, some areas of this zone get little rain, and their climate is semi-arid.

III. Write a note on monsoon.

A monsoon is the system of seasonal winds blowing between 26

degrees north latitude and 23 degrees south latitude on both sides of the equator. The word monsoon comes from the Arabic word mausam (season). In summer, pressure becomes high over the sea and low over the land of the subcontinent, so winds begin to blow from the high-pressure area towards the low-pressure area; coming from the sea, these winds are full of water vapour, rise on reaching land and cause rainfall. In winter the system reverses: pressure becomes low over the sea and high over the land, the winds blow from land towards the sea, and because these winds are dry and cold, the climate becomes dry and cold and there is almost no rain.

Activity (page 46) - Guidance

Guidance: use maps to identify different climatic zones of the world, make a model showing the climatic distribution of Pakistan, and investigate the factors that affect the climate of a region. These are practical activities; the book fixes no single written answer for them.

Chapter 4: Forests of the World

Exercise (pages 62-63): 1. Choose the correct option:

I. The type of forest characterized by extremely cold temperatures and dominated by coniferous trees is

- tropical rainforest
- temperate forest
- ✓ boreal forest (Taiga)
- subtropical forest

II. Forests are considered as the lungs of the Earth because

- ✓ They produce oxygen through photosynthesis.
- They absorb carbon dioxide during transpiration.
- They provide habitat for animals.
- They regulate global temperatures.

III. An example of irrigated plantation is

- Riverine
- Alpine
- Tropical
- ✓ Change Manga

IV. The largest rainforest in the world is:

- Change Manga
- ✓ Amazon forest
- Indus forest

Alpine forest

Exercise: 2. Match the Columns

Match: i. Tropical rainforest - which description fits it?

Located near the equator, high biodiversity. Complete description: tropical rainforests are located near the equator and have high biodiversity.

Match: ii. Boreal forest (Taiga) - which description fits it?

Dominated by coniferous trees, cold climate. Complete description: the boreal forest (taiga) is dominated by coniferous trees and has a cold climate.

Match: iii. Temperate forest - which description fits it?

Experiences four distinct seasons. Complete description: temperate forests experience four distinct seasons.

Match: iv. Subtropical forest - which description fits it?

Difficult to distinguish from tropical rainforests. Complete description: subtropical forests are difficult to distinguish from tropical rainforests.

Exercise: 3. Answer the following short questions:

I. Name any three forest types in the world.

Three forest types in the world are tropical rainforests, temperate forests and boreal forests (taiga). Tropical rainforests grow near the equator, temperate forests grow in North America, Europe and East Asia, and boreal forests stretch across northern Russia, Scandinavia

and Canada.

II. Name any three forests in Pakistan.

Three forests in Pakistan are riverine forests, mangrove forests and alpine forests. Riverine forests, locally called bela, grow along the Indus River and its tributaries in Sindh; mangrove forests grow in coastal and delta regions, including the Indus Delta; and alpine forests occur at high altitudes above the tree line.

III. Define deforestation.

Deforestation is the permanent removal or clearing of forests to use the land for other purposes, such as agriculture, urban development or infrastructure projects. It results in the loss of trees and biodiversity, and harms ecosystems, climate and wildlife.

IV. What is meant by irrigated plantation?

An irrigated plantation is a man-made forest grown in an area where canal water is available. A typical example of an irrigated plantation in Pakistan is Changa Manga, which is a unique man-made forest.

Exercise: 4. Answer the following questions in detail:

I. Discuss different types of forests in the world.

The major forest types of the world are as follows. Tropical rainforests thrive near the equator, where temperatures and humidity are high and rainfall is abundant throughout the year, giving them a great diversity of species. Subtropical forests lie just north and south of the tropical rainforests and are difficult to

distinguish from true rainforests. Temperate forests grace regions like North America, Europe and East Asia and experience four distinct seasons. Boreal forests, also known as taiga, stretch across northern Russia, Scandinavia and Canada, in high northern latitudes with cold climates and long winters, dominated by coniferous trees.

II. Explain the importance of Amazon Rainforest.

The Amazon is the largest rainforest in the world. It produces a significant portion of the oxygen of the world and is often called the lungs of the Earth. Forests absorb carbon from the atmosphere each year, which helps to reduce greenhouse gases and mitigate global climate change, but rampant deforestation releases this stored carbon back into the atmosphere and contributes to global warming. The Amazon faces immense pressure from agriculture, especially beef and soy production, from logging operations, and from population growth, as more people settling in the Amazon basin need land for housing, agriculture and infrastructure, leading to further forest clearance.

III. Write a comprehensive note about different types of forests in Pakistan.

Pakistan has several different kinds of forests. Riverine forests, locally known as bela, thrive along riverbanks, mainly along the Indus River and its major tributaries in Sindh, covering active flood plains. Mangrove forests are located in coastal and delta regions and thrive in both brackish and fresh water, including the Indus Delta. Alpine forests occur at high altitudes, above the tree line at

4000 metres, and are characterized by dwarf shrubs, grasses and hardy plants. Irrigated plantations are man-made forests grown where canal water is available, of which Changa Manga is the typical example in Pakistan. In the northern areas there are coniferous forests, and in Balochistan the Ziarat juniper forest is found. Forests matter for Pakistan because they regulate temperature, humidity and rainfall, take part in the carbon cycle, and significantly affect the economy.

Activity (page 63) - Guidance

Guidance: use maps to identify different forests of the world, make a list of trees in your area, divide the class into two groups to discuss the importance of forests and the dangers of deforestation, and organize a field trip to a nearby forest if possible. These are practical activities; the book fixes no single written answer for them.

Chapter 5: Natural Disasters

Exercise (pages 79-80): 1. Choose the correct answers:

i. Shaking of earth is known as

- flood
- drought
- ✓ earthquake
- wildfires

ii. Water overflowing from its channel due to heavy rainfall is called

- ✓ flood
- drought
- earthquake
- wildfires

iii. Extended dry periods leading to water scarcity is called

- flood
- ✓ drought
- earthquake
- tsunami

iv. 1935 earthquake in Balochistan demolished the city of

- Zhob
- ✓ Quetta
- Khuzdar
- Awaran

v. Cyclones mostly occur in

- plain areas
- dry areas
- mountainous areas
- ✓ coastal areas

Exercise: 2. Fill in the blanks:

i. A natural disaster is any adverse event caused by natural processes that ____ impacts society.

negatively

ii. Developing countries are more ____ to natural disasters.

vulnerable

iii. From 1998-2004, ____ faced one of the worst droughts of its history.

Balochistan

iv. In 2022, intense ____ rains led to disastrous floods in Balochistan.

monsoon

v. Balochistan is highly vulnerable to ____ due to several fault lines.

earthquakes

Exercise: 3. Answer the following short questions:

i. Define natural disasters.

A natural disaster is any adverse event caused by natural processes, such as weather or geologic activity, that negatively impacts human life, property or the environment. Such events can lead to loss of life, destruction of infrastructure and economic setbacks, and they disrupt the smooth functioning of society.

ii. What is earthquake?

An earthquake is the shaking of the earth caused when stress built up along a fault line is released. Several major fault lines run through Pakistan, including the Chaman Fault and the Ornach Fault, which are zones of weakness where stress builds up and is eventually released in the form of earthquakes.

iii. Define flood.

A flood is water overflowing from its channel due to heavy rainfall. Intense monsoon rains can cause widespread flooding, harm homes and infrastructure, and in mountainous terrain the quick runoff can also cause flash floods.

iv. Why developing countries are more vulnerable to disasters?

Developing countries are more vulnerable for several reasons given in the chapter: their buildings often do not follow strict building codes, so they are more likely to collapse in earthquakes or flooding, and poor roads and bridges hinder evacuation and rescue; they lack the financial resources to invest in disaster preparedness; rapid and unplanned urbanization crowds more people into informal settlements that are particularly vulnerable; and environmental degradation, such as deforestation, increases the risk of landslides

and flooding.

v. Mention any three common natural disasters in Balochistan.
Three common natural disasters in Balochistan are earthquakes, droughts and floods. Balochistan is near several active fault lines, including the Chaman and Ornach Faults, making it prone to frequent earthquakes; its arid and semi-arid climate makes it vulnerable to prolonged droughts; and intense monsoon rains have caused disastrous floods, as in 2022, and also in 2010 and 2020. Cyclones are also a danger for its coastal areas near Gwadar and Pasni.

Exercise: 4. Answer the following questions in detail:
i. How does natural disasters affect human life?
Natural disasters affect human life in several ways. They cause loss of life and destruction of homes, businesses, roads and other infrastructure. They cause economic losses: farmers may lose their crops because of too much or too little rain, fishermen can lose their boats in a storm, and small shop owners lose their livelihoods. They displace people from their homes, as in the 2022 Balochistan floods, which displaced thousands of people and caused major damage to agriculture. They also cause environmental degradation, since deforestation increases the risk of landslides and flooding and inadequate land management intensifies the effects of disasters.

ii. How does global warming affect the frequency and intensity of natural disasters? Provide examples.
Global warming increases both the frequency and the intensity of

natural disasters. Greenhouse gas emissions from fossil fuels and deforestation drive climate change, which impacts global weather, sea levels and ecosystems, and climate change drives an increase in the frequency and intensity of disasters. Examples from the chapter are the intense monsoon rains of 2022, which led to disastrous floods in Balochistan, and similar heavy rains in 2010 and 2020, which caused extensive flooding across various districts. Deforestation also intensifies disasters: removing trees reduces the ability of the land to absorb rainfall, leading to more surface runoff and flooding, and because tree roots stabilize soil, their loss makes slopes more vulnerable to landslides.

iii. Describe the natural disasters most common in Balochistan.

The natural disasters most common in Balochistan are earthquakes, droughts, floods and cyclones. Earthquakes are frequent because Balochistan is situated near major fault lines, including the Chaman and Ornach Faults; the major earthquake in Balochistan is the 1935 Quetta earthquake, which demolished the whole city of Quetta and caused widespread destruction and loss of life. Drought is a significant and recurrent problem: from 1998 to 2004 Balochistan faced one of the worst droughts of its history, which ruined lives, destroyed crops, killed a lot of animals and compelled many people to leave their homes. Floods follow intense monsoon rains, as in 2022, when disastrous floods displaced thousands of people and damaged agriculture, and heavy rains in 2010 and 2020 also flooded various districts. Cyclones threaten the coastal areas, particularly near Gwadar and Pasni, where cyclones from the Arabian Sea can cause storm surges and flooding.

Activity (page 80) - Guidance

Guidance: use maps to identify flood and drought prone areas of Balochistan, and enlist the main disasters in your area and identify their causes. These are practical activities depending on your own area; the book fixes no single written answer for them.

Chapter 6: Changing Earth and Human Activities

Exercise (pages 96-97): 1. Choose the correct answers:

I. Types of weathering are

- 2
- ✓ 3
- 4
- 5

II. Weathering due to living organisms is known as

- physical weathering
- chemical weathering
- ✓ biological weathering
- none of these

III. physical weathering is also known as

- chemical weathering
- ✓ mechanical weathering
- biological weathering
- erosion

IV. A distinct landform characterized by steep, straight sides and a narrow valley floor is

- ✓ v-shaped valley
- u-shaped valley
- rapids

glacial valley

V. Accumulations of sand shaped by the wind into various forms are

deflation

✓ dune

mushroom rock

yardang

Exercise: 2. Give short answers of the following questions:

I. Define weathering.

Weathering is the disintegration of rocks into smaller particles. It is the breaking down of rock in place, before any removal or transportation of the material takes place.

II. What is erosion?

Erosion is the natural process of removal and transportation of weathered materials. The main agents of erosion are water, ice and wind, and water and wind are the major factors of erosion.

III. What is chemical weathering?

Chemical weathering is the modification of the chemical composition of rocks and minerals. This kind of breakdown is due to the chemical reaction of the rocks, and it can result in a change in the chemical composition of the residual debris; an example given in the chapter is solution, in which water dissolves limestone or other soluble rocks.

IV. Define river erosion.

River erosion is the process by which rivers and streams wear away their banks and beds, transporting sediment downstream and reshaping the landscape. It is a key part of the fluvial system and shapes river valleys, floodplains and deltas.

V. How Ox-bow lakes are formed?

Ox-bow lakes are formed when a meander is cut off from the main river channel. As the river erodes its banks and deposits sediment, a meander loop becomes separated from the channel, and the cut-off loop remains as a standing body of water shaped like an ox-bow.

Exercise: 3. Answer the following questions in detail:

I. Explain different types of weathering.

There are three types of weathering. Physical or mechanical weathering is the breakdown of rocks without a change in their chemical composition, for example when water enters cracks in rock and freezes, or when rocks are broken by heat and pressure changes. Chemical weathering is the modification of the chemical composition of rocks and minerals by chemical reactions, such as solution, in which water dissolves limestone or other soluble rocks. Biological weathering is weathering due to living organisms, for example when plant roots grow into cracks and break rocks apart, or when animals and microorganisms disturb and break down rock material.

II. Discuss different landforms formed by wind erosion.

Wind erosion creates several landforms. Dunes are accumulations

of sand shaped by the wind into various forms. Mushroom rocks are formed by differential erosion, where the rock at the base erodes more quickly than the rock at the top, often because of variations in rock hardness, leaving a narrow stem and a wider top. Yardangs are streamlined, extended ridges formed by wind abrasion and deflation, typically aligned with the prevailing wind direction. Deflation is the removal of loose, fine-grained particles by the wind, which lowers the ground surface, and a deflation basin (also called a blowout basin) is a depression formed when wind removes fine sediment such as sand or silt from the surface over time.

III. How human activities can contribute to erosion?

Human activities contribute to erosion in several ways described in the chapter. Deforestation removes the vegetation that protects soil, so rainfall runs off the surface instead of being absorbed, increasing flooding and landslides. Agriculture intensifies soil erosion because turning the soil for planting exposes it, leading to soil compaction and erosion. Construction and development create paved surfaces that reduce infiltration and increase surface runoff, while excavation and grading disturb the soil. Mining involves clearing vegetation, removing topsoil and excavating the land, which increases exposure to erosion by wind and water. Poor land management, especially overgrazing, leads to soil compaction and loss of vegetation cover, which increases erosion. Coastal activities, such as building harbours and embankments, alter coastal processes and can increase erosion in adjacent areas.

IV. What are the landforms created by river erosion? Explain.

River erosion creates the following landforms. V-shaped valleys are distinct landforms characterized by steep, straight sides and a narrow valley floor, cut by the river as it wears away its banks and bed. Waterfalls are formed when a river encounters a sudden change in elevation or a resistant rock layer that it cannot easily erode, so the river flows over the edge of a cliff in a vertical drop; rapids occur when a river flows over a shallow gradient or an uneven riverbed. Ox-bow lakes are formed when a meander is cut off from the main river channel. Through deposition as well as erosion, rivers also shape floodplains and deltas, where the sediment they carry is laid down.

Activity (page 97) - Guidance

Guidance: find out which types of weathering occur in your area and how human activities are changing the landforms in your area, and make a model of landforms made by a river. These are practical activities depending on your own area; the book fixes no single written answer for them.